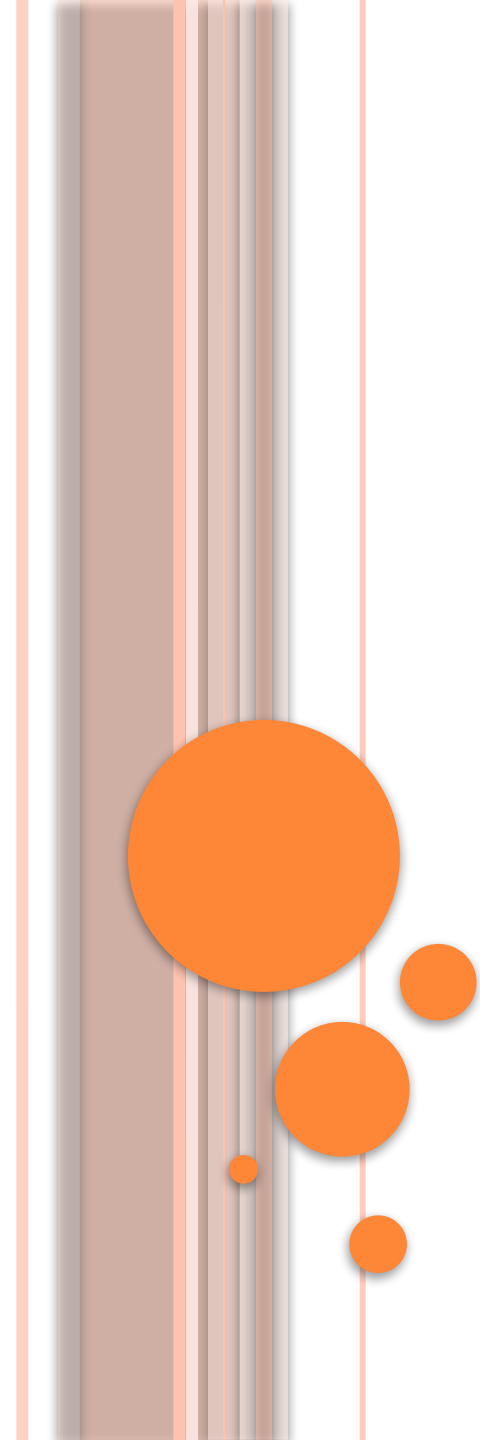


A decorative graphic on the left side of the slide. It features a large orange circle at the top left, with several smaller orange circles of varying sizes arranged in a cluster below it. To the left of these circles are several vertical lines of different colors, including brown, orange, and grey, some of which are thicker than others.

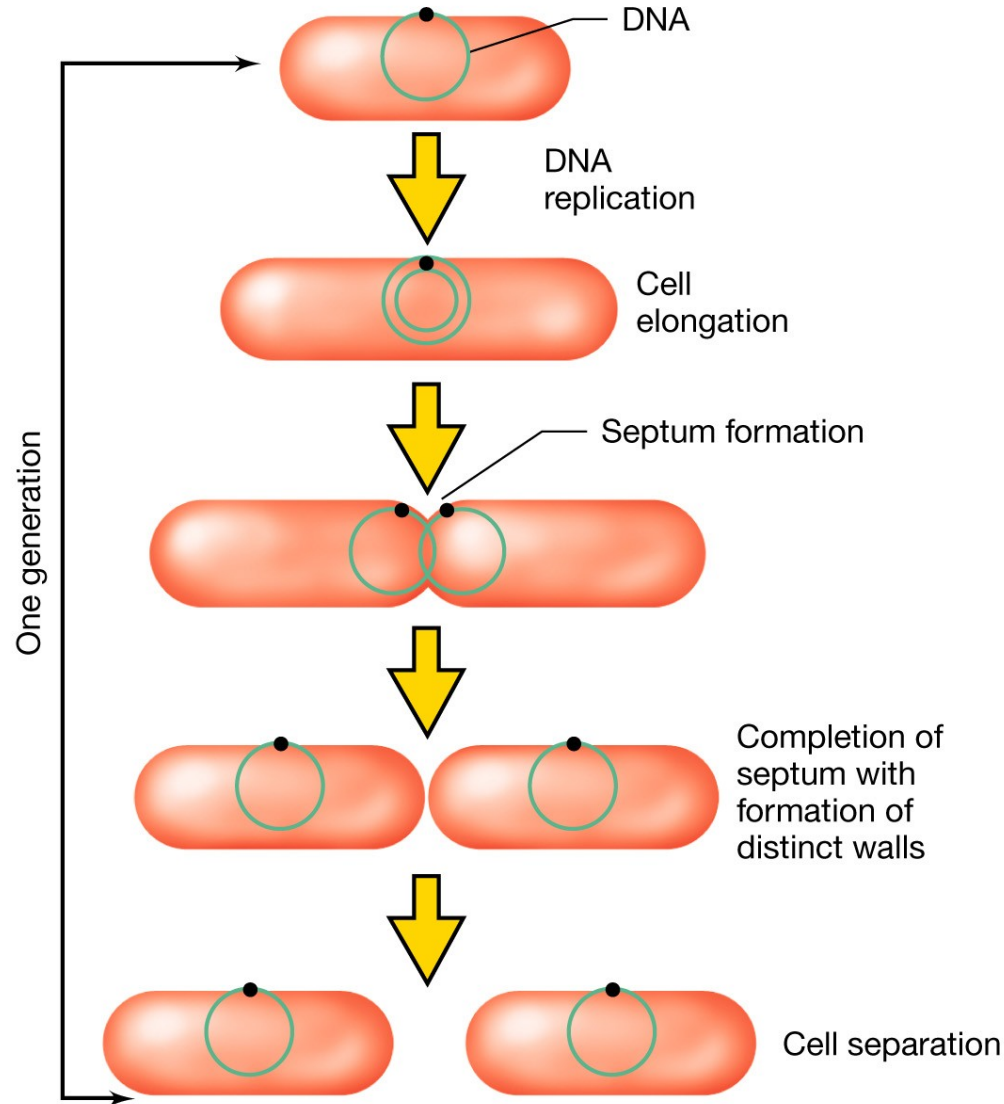
MICROBIAL GROWTH

BACTERIAL GROWTH - INCREASE IN THE NUMBER OF CELLS

- Binary Fission
- Generation Time (Doubling Time)
 - time required for a cell to divide
 - most about 1 Hr. To 3 Hrs.
 - *E. coli* - 20 minutes
 - *Mycobacterium tuberculosis* - 24 Hrs.



BACTERIAL GROWTH – BINARY FISSION



BINARY FISSION - UNCHECKED

- *E. coli* - generation time of 20 min.
- 20 generations (about 7 hrs.)
 - 1 million cells
- 30 generations (about 10 hrs.)
 - 1 billion cells
- 72 generations (about 24 hrs.)
 - 1×10^{21}
 - 1,000,000,000,000,000,000,000 cells



LIMITING FACTORS IN THE ENVIRONMENT

- Lack of food, water or nutrients
- space
- accumulation of metabolic wastes
- lack of oxygen
- changes in pH
- temperature



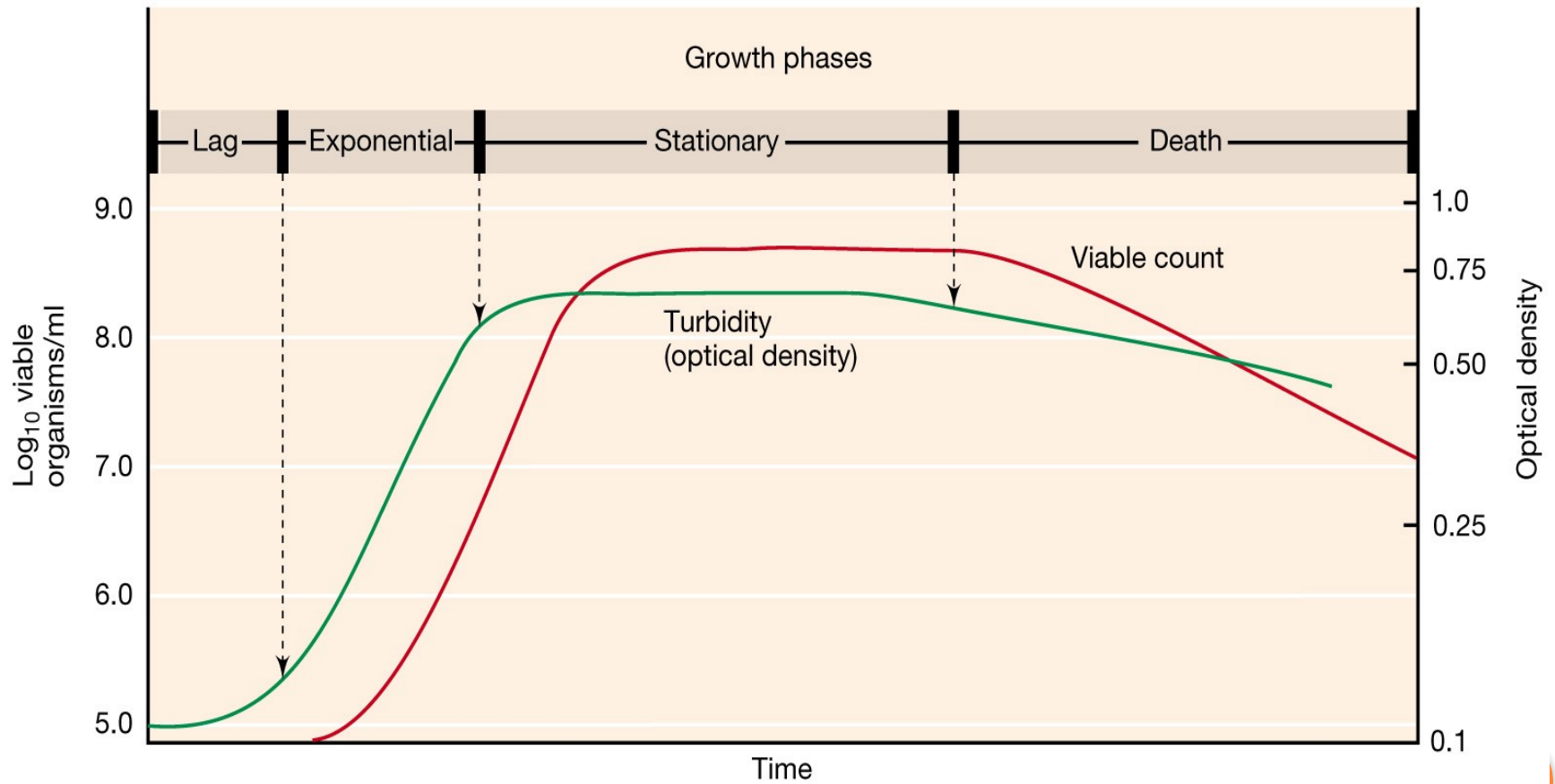
BACTERIAL GROWTH

- Exponential growth via **binary fission**
- Generation or **doubling time** (from 20 min to 24 h)
- **Phases of growth**
 - Lag phase
 - Exponential (logarithmic [log]) phase
 - Stationary phase
 - Death phase

Compare growth in liquid and on solid media



BACTERIAL GROWTH CURVE IN BATCH CULTURE

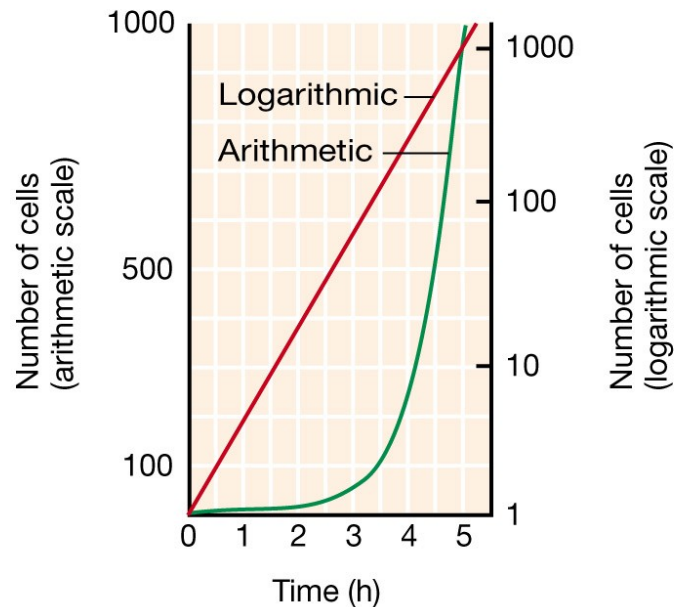


Exponential growth = balanced growth

BACTERIAL GROWTH RATE

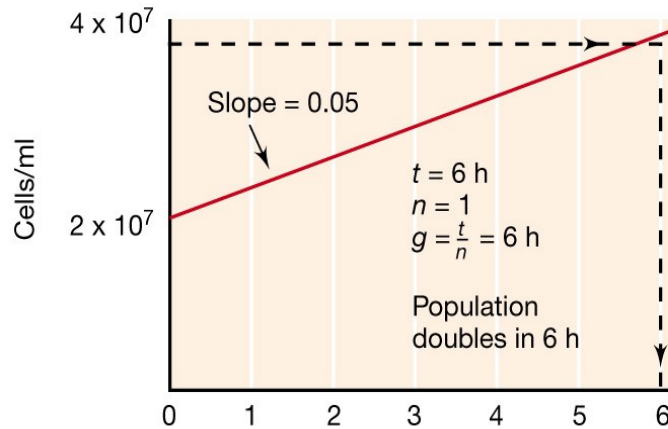
Time (h)	Total number of cells	Time (h)	Total number of cells
0	1	4	256
0.5	2	4.5	512
1	4	5	1,024
1.5	8	5.5	2,048
2	16	6	4,096
2.5	32	.	.
3	64	.	.
3.5	128	10	1,048,576

(a)

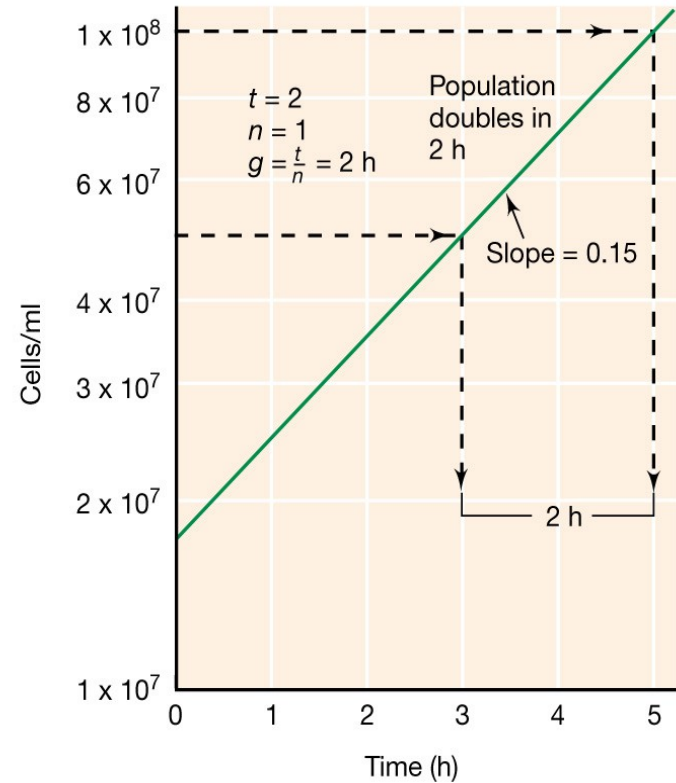


(b)

Calculating growth rate



(a)



(b)

Generation time (g) = t/n ; where t = elapsed time (hr) and n = number of generations.



GENERATION TIMES

Bacterium	Medium	Generation Time (minutes)
<i>Escherichia coli</i>	Glucose-salts	17
<i>Bacillus megaterium</i>	Sucrose-salts	25
<i>Streptococcus lactis</i>	Milk	26
<i>Streptococcus lactis</i>	Lactose broth	48
<i>Staphylococcus aureus</i>	Heart infusion broth	27-30
<i>Lactobacillus acidophilus</i>	Milk	66-87
<i>Rhizobium japonicum</i>	Mannitol-salts-yeast extract	344-461
<i>Mycobacterium tuberculosis</i>	Synthetic	792-932
<i>Treponema pallidum</i>	Rabbit testes	1980



PHASES OF GROWTH

- 4 Phases
- 1. Lag Phase
- 2. Log Phase
- 3. Stationary Phase
- 4. Death Phase



1. LAG PHASE

- Bacteria are first introduced into an environment or media
- Bacteria are “checking out” their surroundings
- cells are very active metabolically
- # of cells changes very little
- 1 hour to several days



2. LOG PHASE

- Rapid cell growth (exponential growth)
- population **doubles** every generation
- microbes are sensitive to adverse conditions
 - antibiotics
 - anti-microbial agents



3. STATIONARY PHASE

- Death rate = rate of reproduction
- cells begin to encounter environmental stress
 - lack of nutrients
 - lack of water
 - not enough space
 - metabolic wastes
 - oxygen
 - pH

Endospores would form now



4. DEATH PHASE

- Death rate $>$ rate of reproduction
- Due to limiting factors in the environment



LICHENS

- Leafy or encrusting microbial symbiosis
- Often found growing on bare rocks, tree trunks, house roofs, and the surfaces of bare soils
- Consist of a mutualistic relationship between a fungus and an alga (or cyanobacterium)
 - Alga is photosynthetic and produces organic matter for the fungus
 - The fungus provides a structure within which the phototrophic partner can grow protected from erosion by rain or wind



Lichens



T. D. Brock

(a)

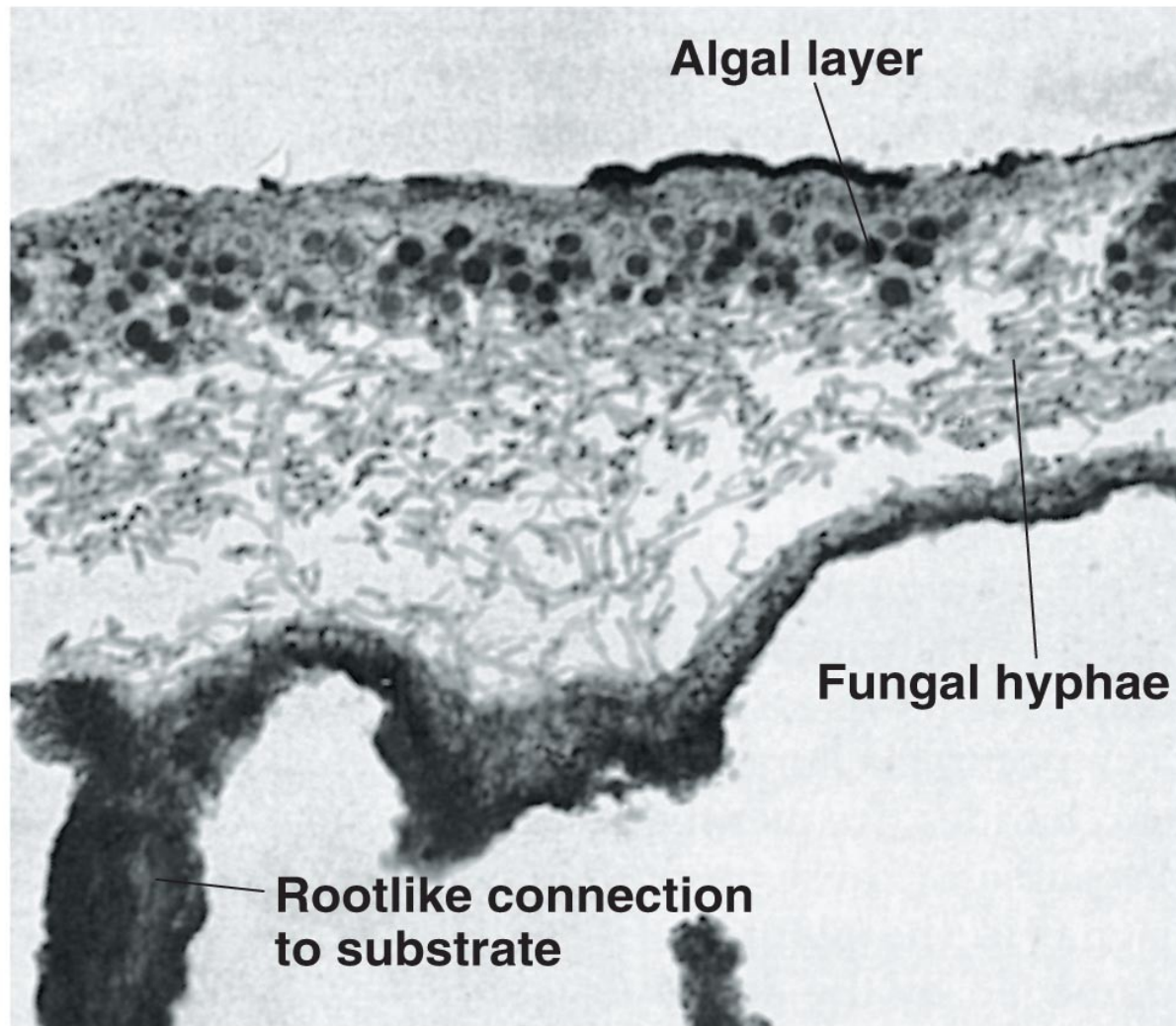


M. T. Madigan

(b)



Lichen Structure



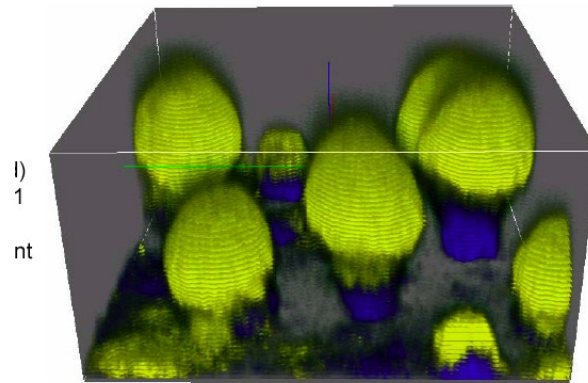
T. D. Brock



BACTERIA AS COMMUNITIES

“SOCIOMICROBIOLOGY”

- Biofilms
- Quorum sensing



BIOFILMS: MICROBIAL GROWTH ON SURFACES

Surfaces are important microbial habitats
because

- Nutrients adsorb to surfaces

- Microbial cells can attach to surfaces



BIOFILMS: MICROBIAL GROWTH ON SURFACES

Biofilms

Assemblages of bacterial cells adhered to a surface and enclosed in an adhesive matrix excreted by the cells

The matrix is typically a mixture of polysaccharides

Biofilms trap nutrients for microbial growth and help prevent detachment of cells in flowing systems



BIOFILMS

"COMMUNITY OF MICROBES EMBEDDED IN AN ORGANIC POLYMER MATRIX THAT ADHERES TO A SURFACE"

- 3D structure containing 1 or more microbial species
 - Forms at interfaces
 - Solid/liquid, liquid/air, air/solid
 - Spatial heterogeneity
 - Exopolysaccharide "glue"
- Important in human infections
 - Decreased antibiotic susceptibility
 - Less conspicuous to immune system
- Targets organisms to favorable niches
- Protection from physical injury
- Promote horizontal gene transfer
- Enhance cell-cell communication

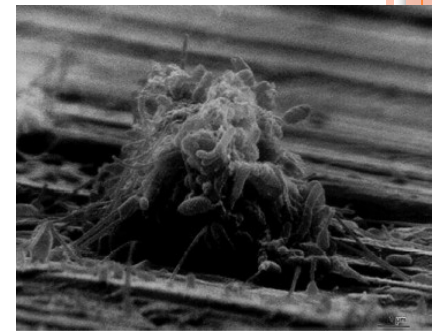
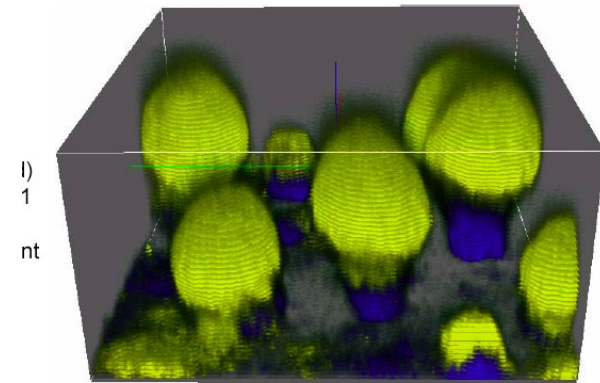
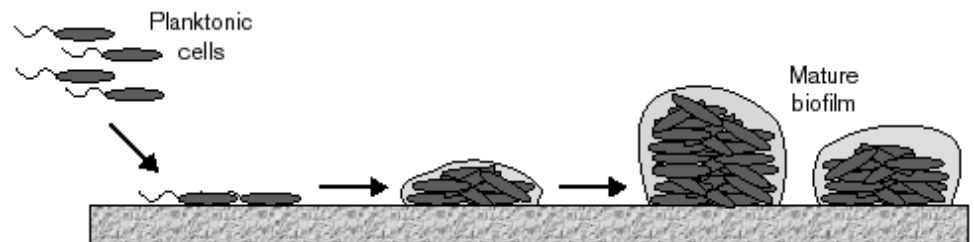


FIG. 1. Scanning electron micrograph of a biofilm on a metal surface from an industrial water system.

COMPONENTS

- **Microorganisms**
 - Single or heterogeneous
 - Sessile (attached) vs planktonic (unattached)
- **Glycocalyx**
 - “Glue” containing polysaccharide
 - Traps/concentrates nutrients
- **Surface**
 - Abiotic (inert)
 - Biotic (living)



BIOFILMS

- Abundant in aquatic ecosystems
- Targets specialized microbes
- Nutritionally favorable, abundant nutrients
 - Symbiosis
 - Rhizobium/legumes
 - Ruminant digestive tracts: degrade/recycle insoluble materials
 - Non-hostile environment, Protection from predators
- Insurance hypothesis
 - Presence of diverse subpopulations increases survival of community as a whole

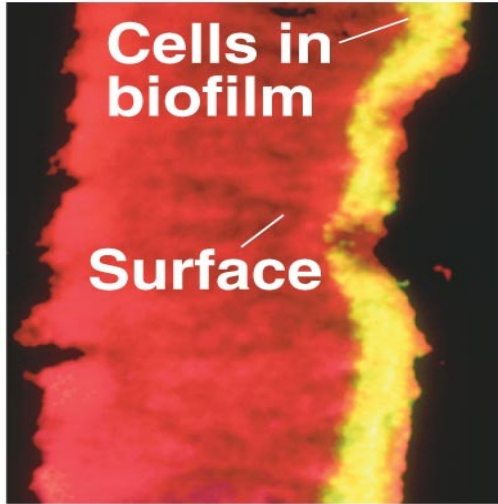


BACTERIAL SPECIES ASSOCIATED WITH BIOFILMS

- *Pseudomonas aeruginosa*
- *Staphylococcus epidermidis*
- *Staphylococcus aureus*
- *Escherichia coli* O157:H7
- *Lactobacillus* spp.
- Streptococcal spp.
- *Salmonella*
- And
Yeast
- *Candida albicans*



Examples of Microbial Biofilms

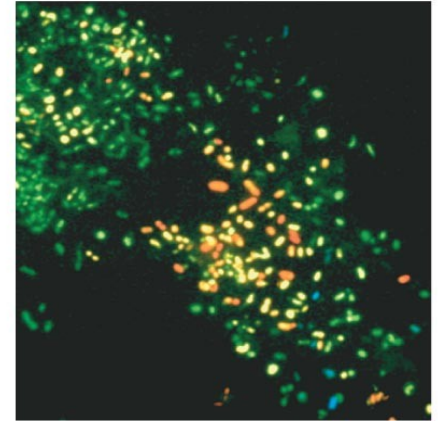


C.-T. Huang, Karen Xu, Gordon McFeters, and Philip S. Stewart

(a) Cross-Sectional View of Experimental Biofilm



Biofilm on medical catheter



Cindy E. Morris

(b)

Natural Biofilm on a Leaf Surface

Biofilm of Iron-Oxidizing Prokaryotes Attached to Rocks



J.M. Sánchez, J.J. deLope, and Ricardo Amils

(c)

BIOFILMS: ADVANTAGES AND CONTROL

Bacteria form biofilms for several reasons

Self-defense

Biofilms resist physical forces that sweep away unattached cells, phagocytosis by immune system cells, and penetration of toxins (e.g., antibiotics)

Allows cells to remain in a favorable niche

Allows bacterial cells to live in close association with one another

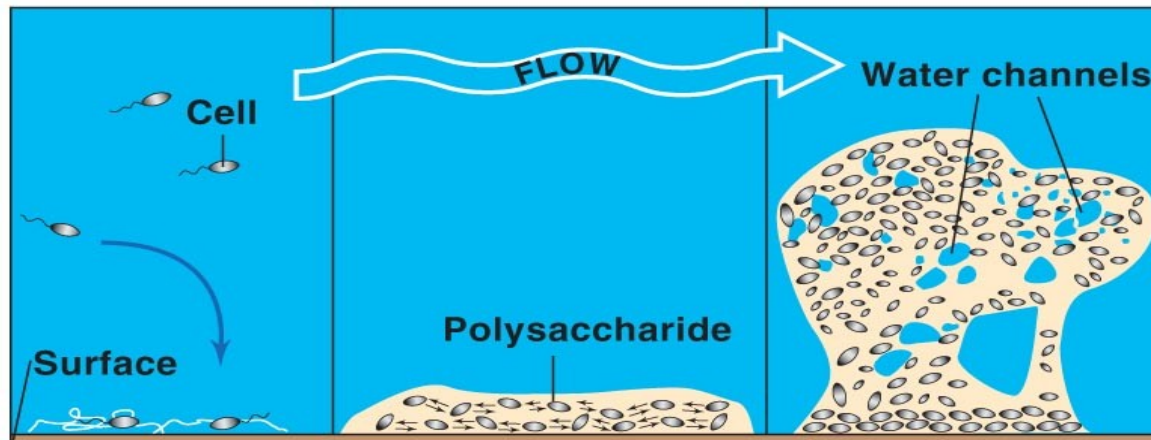


Biofilm Formation

Attachment
(adhesion of a few cells to a suitable solid surface)

Colonization
(intercellular communication, growth and polysaccharide formation)

Development
(more growth and polysaccharide)



(a)



BIOFILM FORMATION

- Stepwise process
 - Formation of conditioning layer
 - Equipment design and condition
 - Organic (proteins) or inorganic matter
 - pH extremes, protein denaturation
 - Extracellular polysaccharide production
 - Varies based on bacterial species
 - Bacterial adhesion to conditioning layer
 - fimbriae, pili, flagella
 - electrostatic, hydrogen, covalent bonds
 - hydrophobic, Van der Waals interactions.



BIOFILM FORMATION

- Rapid development with nutrient source
 - 1 hour-10% of bacteria - irreversible adhesion
 - 8 hours-91% of bacteria - irreversible adhesion



PLACES OF BIOFILM FORMATION

- Extended run operations
- Dry cleaning only during week
- Cleaned daily, but not with stringent regime to remove biofilms
 - Floors
- Not cleaned every 24 hours
 - Walls
 - Drains



MATURE BIOFILMS

Mechanical action and strong chemicals applied for lengthy contact times and high temperatures are necessary for removal

Prior to a biofilm becoming mature, shear flow force may promote removal



QUORUM SENSING

- Intermicrobe communication
 - Signaling (language) system that allows bacteria to sense cell density
- Inter- vs intra-species
- Early examples
 - *Vibrio harveyi*-only produces light at high cell densities
 - Can induce light by adding supernatants from stationary phase cultures
 - *Strep pneumoniae*-competence induced at late-log phase
 - Produces “competence” factor at high cell densities
- Important in many processes
 - Symbiosis
 - Virulence
 - Antibiotic production
 - Biofilm formation
- Microbe-host communication



ATP BIOLUMINESCENCE

- 1.0×10^3 CFU/g needed for detection
 - Actively growing cells
 - Produce higher levels of ATP
- Bacteria embedded in biofilms
 - Less movement = less energy required
 - Less energy required = less ATP generated
 - Greater than 1.0×10^3 CFU/g needed for ATP detection



REMOVAL OF BIOFILMS: REMOVAL WITH CHEMICALS

- Cleaning and sanitation agents
 - Formulation and concentration
 - 23% hydrogen peroxide and 4% peracetic acid used at 1-2% working concentration
 - Any oxidative agent
 - Acids will remove rust, inorganic soil
- New technologies - accelerated hydrogen peroxide Chlorinated alkaline (0.5 oz/gal), pH 11-12
 - 160°F circulating water for 20 minutes
- Phosphoric acid (1oz/gal)
 - 160°F circulating water for 20 minutes
- Chlorine solution (0.3 oz/gal)
 - 160°F circulating water for 15 minutes



CONDITIONS FOR REMOVAL

- Time of chemical application
 - Increased contact time of cleaners/sanitizers will yield better results
 - At least 5 minutes for hydrogen peroxide and acids

Temperature

Use hot (130 °F) water

- Agitation or mechanical action
 - Use of brushes/cleaning tools
 - Circulating water
 - Water used under pressure

